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Université: Institut Universitaire des Sciences (IUS)

TD N° 2: Administration Systèmes

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Objectif

L'Objectif de ce travail dirigé est de mettre en place une solution de sauvegarde sous Linux afin de protéger les données importantes d'un serveur. Il vise à apprendre à utiliser des outils comme **rsync**, **SSH** et **cron** pour réaliser des sauvegardes locales et distantes, les automatiser et assurer une restauration rapide des fichiers en cas de perte ou d'incident.

Exercice 1

1. Strategie de sauvegarde

Données à sauvegarder:

/var/www/

/etc/

/home/devops/projects

Dossier de sauvegarde:

- /backup

1.1 Créer le script

```
sudo mkdir /backup
```

```
mkdir -p /home /devops/projects
```

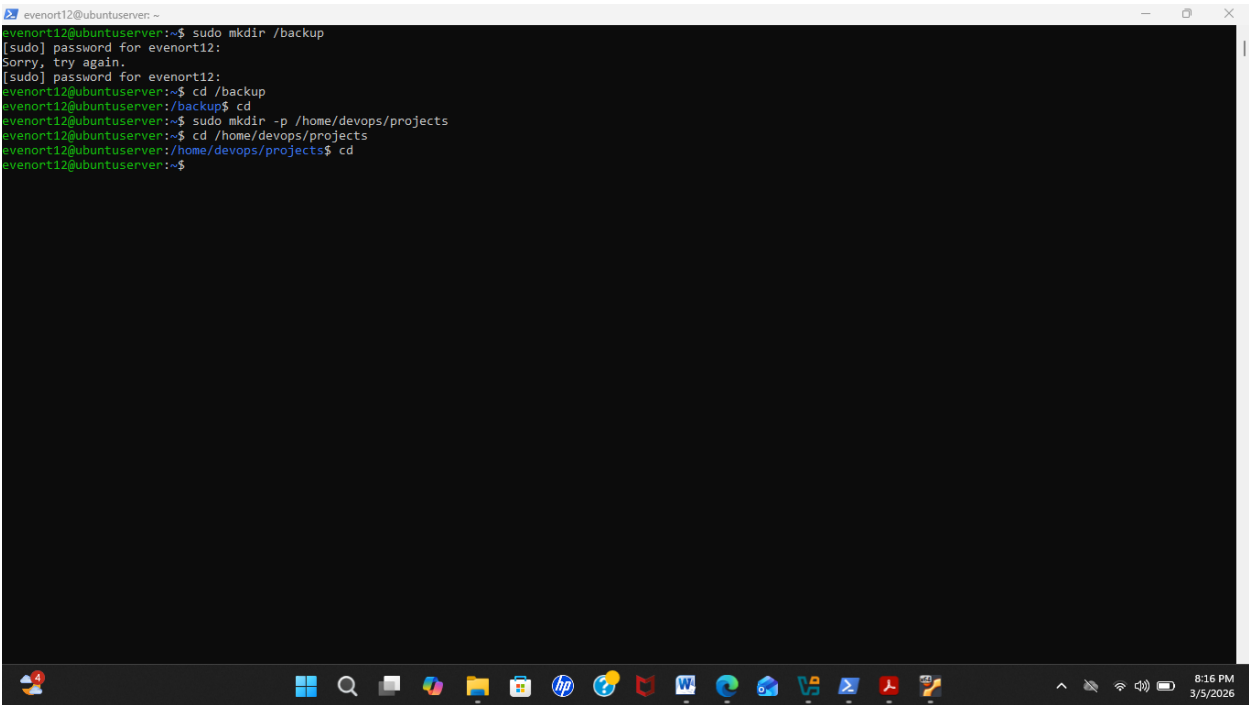


Figure 1 Cette image illustre que le script a été créé avec succès.

1.2 Vérification depuis le serveur

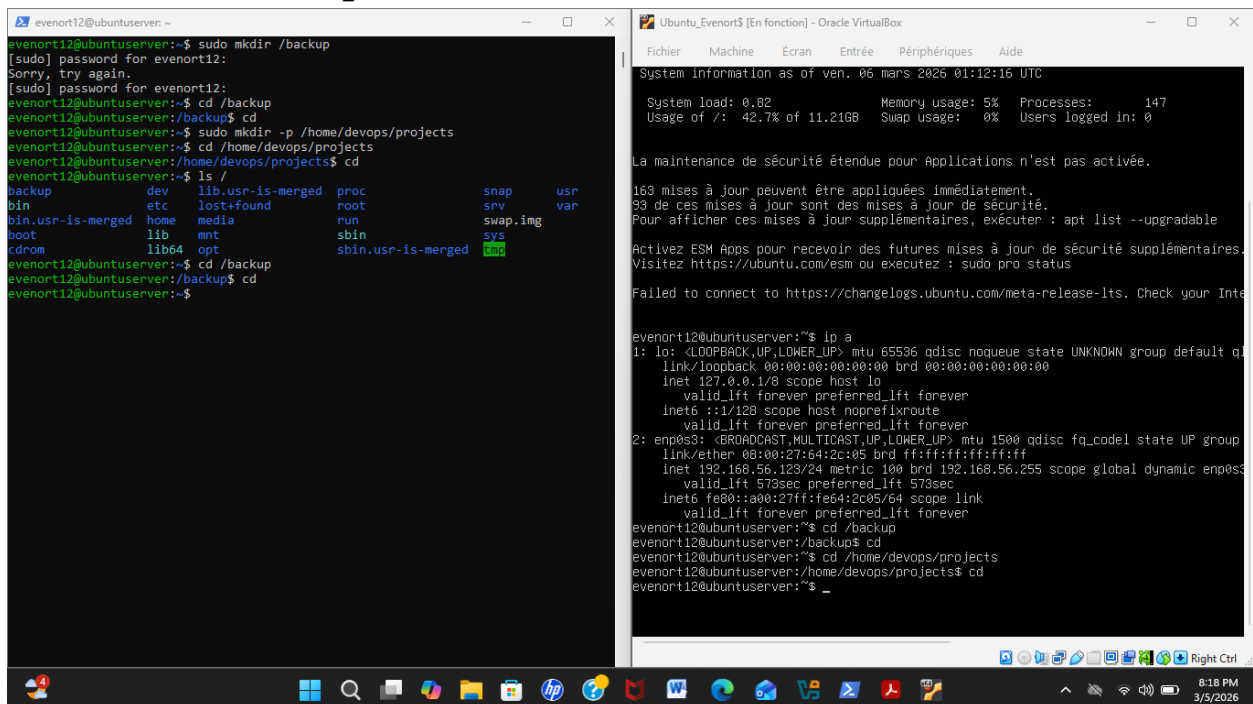
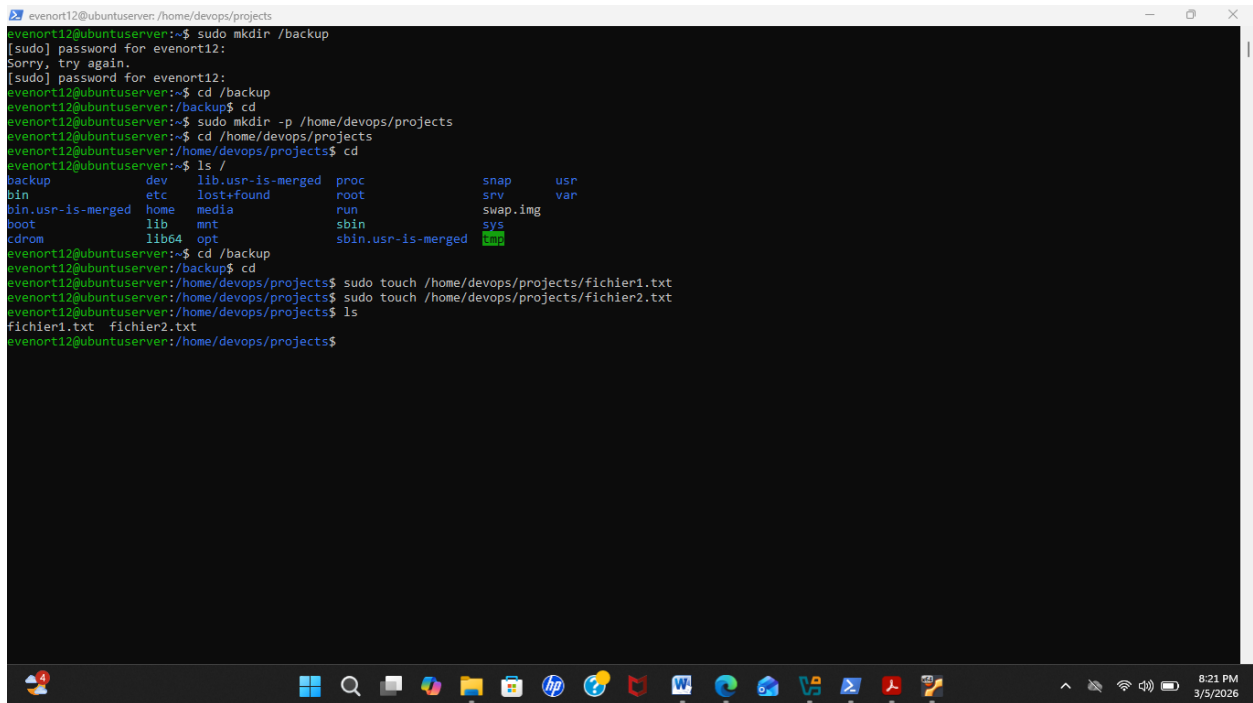


Figure 2 : Cette image montre que le test de vérification a été réalisé avec succès.

2. Créer des fichiers de test

Sudo touch /home/devops/projects /fichier1.txt

A terminal window titled 'evenort12@ubuntu: /home/devops/projects' showing a series of commands and their outputs. The user creates a 'backup' directory, then a 'projects' directory under '/home/devops/'. After a directory listing, the user uses 'sudo touch' to create 'fichier1.txt' and 'fichier2.txt' in the 'projects' directory. The final 'ls' command confirms the files' existence.

```
evenort12@ubuntu: /home/devops/projects
evenort12@ubuntu:~$ sudo mkdir /backup
[sudo] password for evenort12:
Sorry, try again.
[sudo] password for evenort12:
evenort12@ubuntu:~$ cd /backup
evenort12@ubuntu:~/backup$ cd
evenort12@ubuntu:~$ sudo mkdir -p /home/devops/projects
evenort12@ubuntu:~$ cd /home/devops/projects
evenort12@ubuntu:~/devops/projects$ cd
evenort12@ubuntu:~/devops/projects$ ls /
backup      dev          lib.usr-is-merged  proc          snap          usr
bin         etc          lost+found         root          srv           var
bin.usr-is-merged  home        media             run          swap.img
boot        lib          mnt              sbin         sys
cdrom       lib64       opt             sbin.usr-is-merged  sys

evenort12@ubuntu:~$ cd /backup
evenort12@ubuntu:~/backup$ cd
evenort12@ubuntu:~/devops/projects$ sudo touch /home/devops/projects/fichier1.txt
evenort12@ubuntu:~/devops/projects$ sudo touch /home/devops/projects/fichier2.txt
evenort12@ubuntu:~/devops/projects$ ls
fichier1.txt  fichier2.txt
evenort12@ubuntu:~/devops/projects$
```

Figure 3: Cette image montre que les fichiers de test ont été créés avec succès.

2.1 Ajouter du contenu:

sudo nano fichier1.txt

sudo nano fichier2.txt

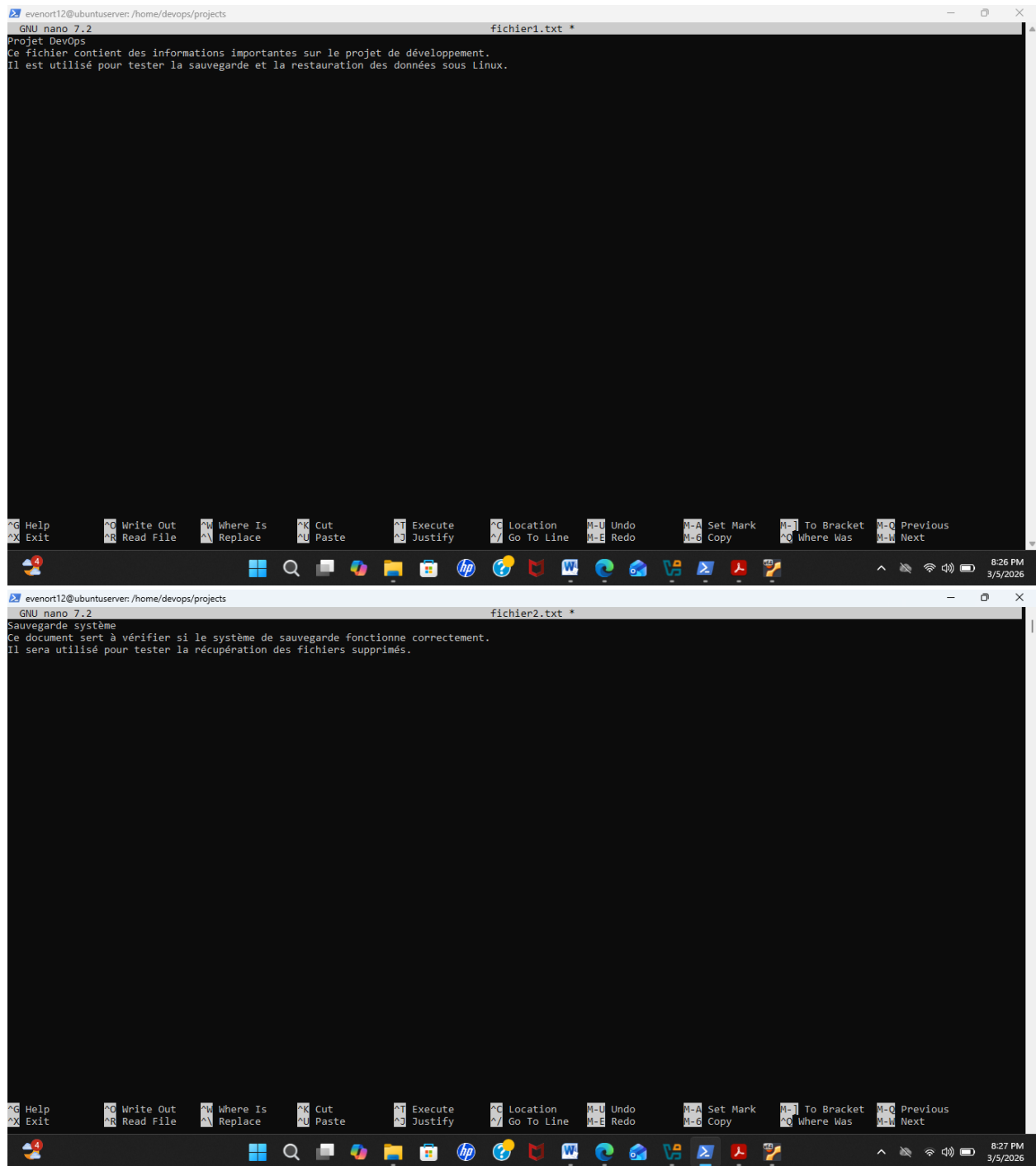


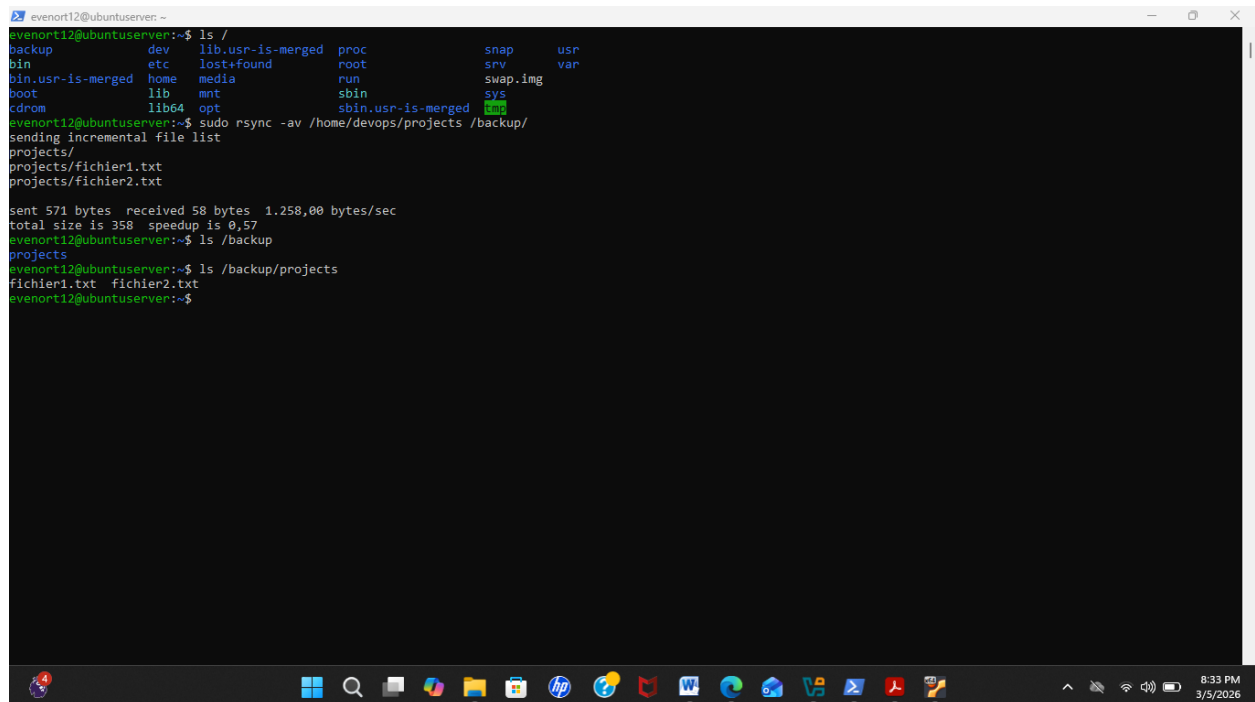
Figure 4, 5: Ces images attestent que les contenus ont été ajoutés correctement.

```
evenort12@ubuntu: /home/devops/projects
evenort12@ubuntu:~$ sudo mkdir /backup
[sudo] password for evenort12:
Sorry, try again.
[sudo] password for evenort12:
evenort12@ubuntu:~$ cd /backup
evenort12@ubuntu:/backup$ cd
evenort12@ubuntu:~$ sudo mkdir -p /home/devops/projects
evenort12@ubuntu:~$ cd /home/devops/projects
evenort12@ubuntu:/home/devops/projects$ cd
evenort12@ubuntu:/home/devops/projects$ ls /
backup      dev          lib.usr-is-merged  proc          snap         usr
bin         etc          lost+found         root          srv          var
bin.usr-is-merged  home        media             run          swap.img
boot        lib          mnt              sbin         sys
cdrom       lib64       opt             sbin.usr-is-merged  sys

evenort12@ubuntu:~$ cd /backup
evenort12@ubuntu:/backup$ cd
evenort12@ubuntu:/home/devops/projects$ sudo touch /home/devops/projects/fichier1.txt
evenort12@ubuntu:/home/devops/projects$ sudo touch /home/devops/projects/fichier2.txt
evenort12@ubuntu:/home/devops/projects$ ls
fichier1.txt  fichier2.txt
evenort12@ubuntu:/home/devops/projects$ sudo nano fichier1.txt
evenort12@ubuntu:/home/devops/projects$ sudo nano fichier2.txt
evenort12@ubuntu:/home/devops/projects$ sudo cat fichier1.txt
Projet DevOps
Ce fichier contient des informations importantes sur le projet de développement.
Il est utilisé pour tester la sauvegarde et la restauration des données sous Linux.
evenort12@ubuntu:/home/devops/projects$ sudo cat fichier2.txt
Sauvegarde système
Ce document sert à vérifier si le système de sauvegarde fonctionne correctement.
Il sera utilisé pour tester la récupération des fichiers supprimés.
evenort12@ubuntu:/home/devops/projects$
```

Figure 6 : Cette image montre les lignes de commande utilisées pour effectuer l'opération.

3. Création de la sauvegarde pour les dossiers



```
evenort12@ubuntu:~$ ls /
backup      dev         lib.usr-is-merged  proc        snap        usr
bin         etc         lost+found         root        srv          var
bin.usr-is-merged  home       media              run         swap.img
boot        lib         mnt                sbin        sys
cdrom       lib64       opt                sbin.usr-is-merged  sys

evenort12@ubuntu:~$ sudo rsync -av /home/devops/projects /backup/
sending incremental file list
projects/
projects/fichier1.txt
projects/fichier2.txt

sent 571 bytes  received 58 bytes  1.258,00 bytes/sec
total size is 358  speedup is 0,57
evenort12@ubuntu:~$ ls /backup
projects
evenort12@ubuntu:~$ ls /backup/projects
fichier1.txt  fichier2.txt
evenort12@ubuntu:~$
```

Figure 7: Cette image montre que la sauvegarde des dossiers sur le serveur a été créée avec succès.

4 – Test restauration

Sudo rm /home/devops/projects/fichier1.txt

4.1 Restauration du fichier

Sudo rsync -av /backup/projects/ /home/devops/projects/

```
evenort12@ubuntu:~$ rm /home/devops/projects/fichier1.txt
rm: remove write-protected regular empty file '/home/devops/projects/fichier1.txt'?
evenort12@ubuntu:~$ ls /home/devops/projects
fichier1.txt  fichier2.txt
evenort12@ubuntu:~$ ls -l
total 8
-rw-r--r-- 1 root root 182 mars  5 16:16 fichier1.txt
-rw-r--r-- 1 root root 176 mars  5 16:19 fichier2.txt
evenort12@ubuntu:~$ sudo rm /home/devops/projects/fichier1.txt
[sudo] password for evenort12:
evenort12@ubuntu:~$ ls /home/devops/projects
fichier2.txt
evenort12@ubuntu:~$ rsync -av /backup/projects/ /home/devops/projects
sending incremental file list
rsync: [generator] failed to set times on "/home/devops/projects/.": Operation not permitted (1)
./
fichier1.txt
rsync: [receiver] mkstemp "/home/devops/projects/.fichier1.txt.fIjLnU" failed: Permission denied (13)

sent 149 bytes  received 38 bytes  374,00 bytes/sec
total size is 0  speedup is 0,00
rsync error: some files/attrs were not transferred (see previous errors) (code 23) at main.c(1338) [sender=3.2.7]
evenort12@ubuntu:~$ sudo rsync -av /backup/projects/ /home/devops/projects
sending incremental file list
./
fichier1.txt

sent 149 bytes  received 38 bytes  374,00 bytes/sec
total size is 0  speedup is 0,00
evenort12@ubuntu:~$
```

Figure 8: Cette image montre que le test de restauration du fichier a été réalisé avec succès.

4.2 Vérifiez à nouveau: ls /home/devops/projects

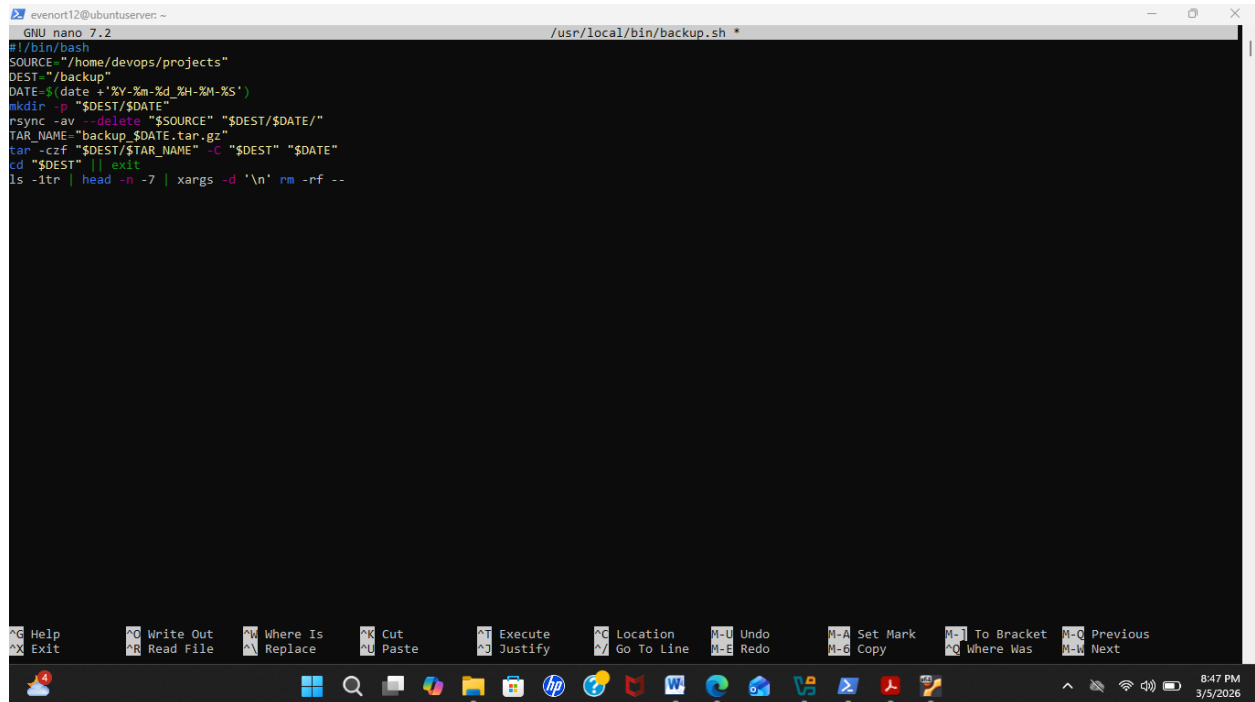
```
evenort12@ubuntu:~$ ls /home/devops/projects
fichier2.txt
evenort12@ubuntu:~$ sudo rsync -av /backup/projects/ /home/devops/projects
sending incremental file list
./
fichier1.txt

sent 334 bytes  received 38 bytes  744,00 bytes/sec
total size is 350  speedup is 0,96
evenort12@ubuntu:~$ ls /home/devops/projects
fichier1.txt  fichier2.txt
evenort12@ubuntu:~$
```

Figure 9: Cette image montre que la sauvegarde et la restauration des fichiers fonctionnent correctement sur le serveur.

4.3 Ajouter du contenu:

`sudo nano /usr/local/bin/backup.sh`



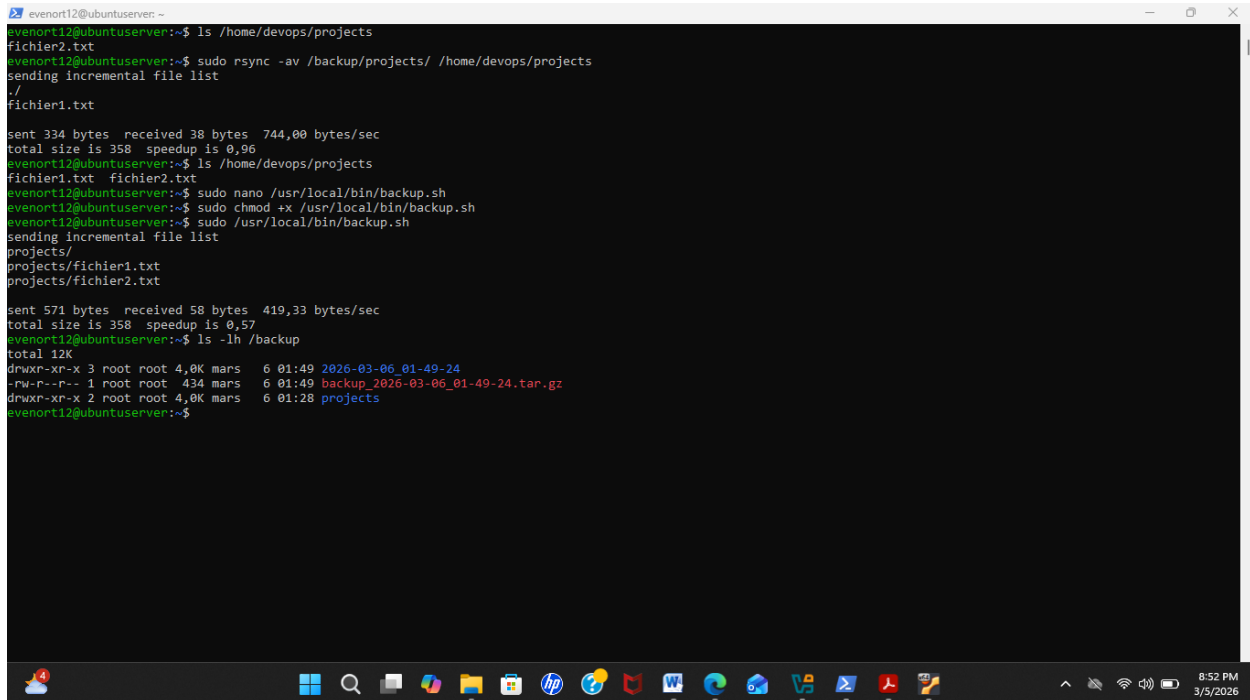
```
evenort12@ubuntu: ~  
GNU nano 7.2 /usr/local/bin/backup.sh *  
#!/bin/bash  
SOURCE="/home/devops/projects"  
DEST="/backup"  
DATE=$(date +%Y-%m-%d_%H-%M-%S)  
mkdir -p "$DEST/$DATE"  
rsync -av --delete "$SOURCE" "$DEST/$DATE/"  
TAR_NAME="backup_$DATE.tar.gz"  
tar -czf "$DEST/$TAR_NAME" -C "$DEST" "$DATE"  
cd "$DEST" || exit  
ls -ltr | head -n -7 | xargs -d '\n' rm -rf --
```

Figure 10: Ces images montrent que les contenus ont été ajoutés correctement.

4.4 Rendre le script executable: `sudo chmod +x /usr/local/bin/backup.sh`

Tester le script: `sudo /usr/local/bin/backup.sh`

Vérification des résultats :: `ls -lh /backup`



```
evenort12@ubuntu:~$ ls /home/devops/projects
fichier2.txt
evenort12@ubuntu:~$ sudo rsync -av /backup/projects/ /home/devops/projects
sending incremental file list
./
fichier1.txt

sent 334 bytes  received 38 bytes  744,00 bytes/sec
total size is 358  speedup is 0,96
evenort12@ubuntu:~$ ls /home/devops/projects
fichier1.txt  fichier2.txt
evenort12@ubuntu:~$ sudo nano /usr/local/bin/backup.sh
evenort12@ubuntu:~$ sudo chmod +x /usr/local/bin/backup.sh
evenort12@ubuntu:~$ sudo /usr/local/bin/backup.sh
sending incremental file list
projects/
projects/fichier1.txt
projects/fichier2.txt

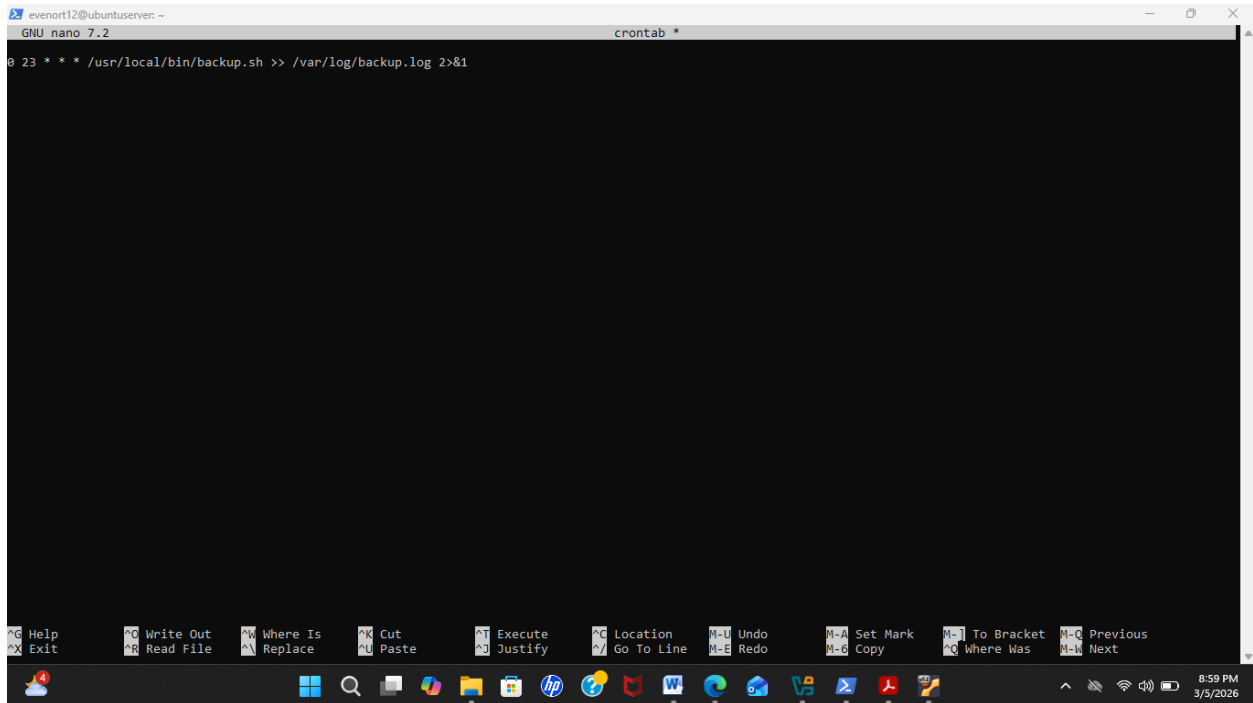
sent 571 bytes  received 58 bytes  419,33 bytes/sec
total size is 358  speedup is 0,57
evenort12@ubuntu:~$ ls -lh /backup
total 12K
drwxr-xr-x 3 root root 4,0K mars  6 01:49 2026-03-06_01-49-24
-rw-r--r-- 1 root root 434 mars  6 01:49 backup_2026-03-06_01-49-24.tar.gz
drwxr-xr-x 2 root root 4,0K mars  6 01:28 projects
evenort12@ubuntu:~$
```

Figure 11: Sur cette image, la commande `ls -lh /backup` permet de vérifier les fichiers sauvegardés et d'afficher leurs détails (taille, permissions, date de modification).

5. Automatisation avec cron

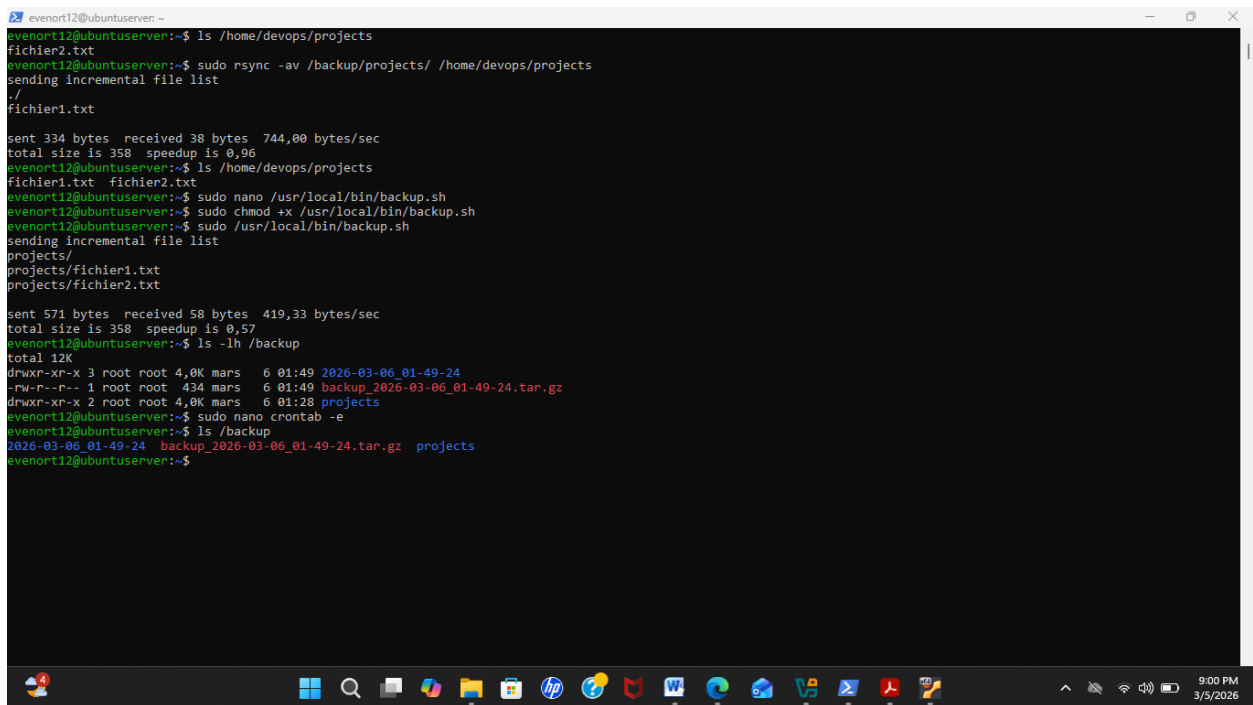
`Sudo nano crontab -e`

- **Ajouter:** `0 23 * * * /usr/local/bin/backup.sh`



The screenshot shows a terminal window with the title 'evenort12@ubuntu: ~'. The nano editor is open, editing a file named 'crontab'. The content of the file is: `0 23 * * * /usr/local/bin/backup.sh >> /var/log/backup.log 2>&1`. The nano editor's status bar at the bottom shows 'GNU nano 7.2'. The system's taskbar is visible at the bottom of the window, showing various application icons and the system clock indicating 8:59 PM on 3/5/2026.

Figure 12: Sur cette image, le contenu du fichier a été ajouté afin d'automatiser la sauvegarde avec cron.

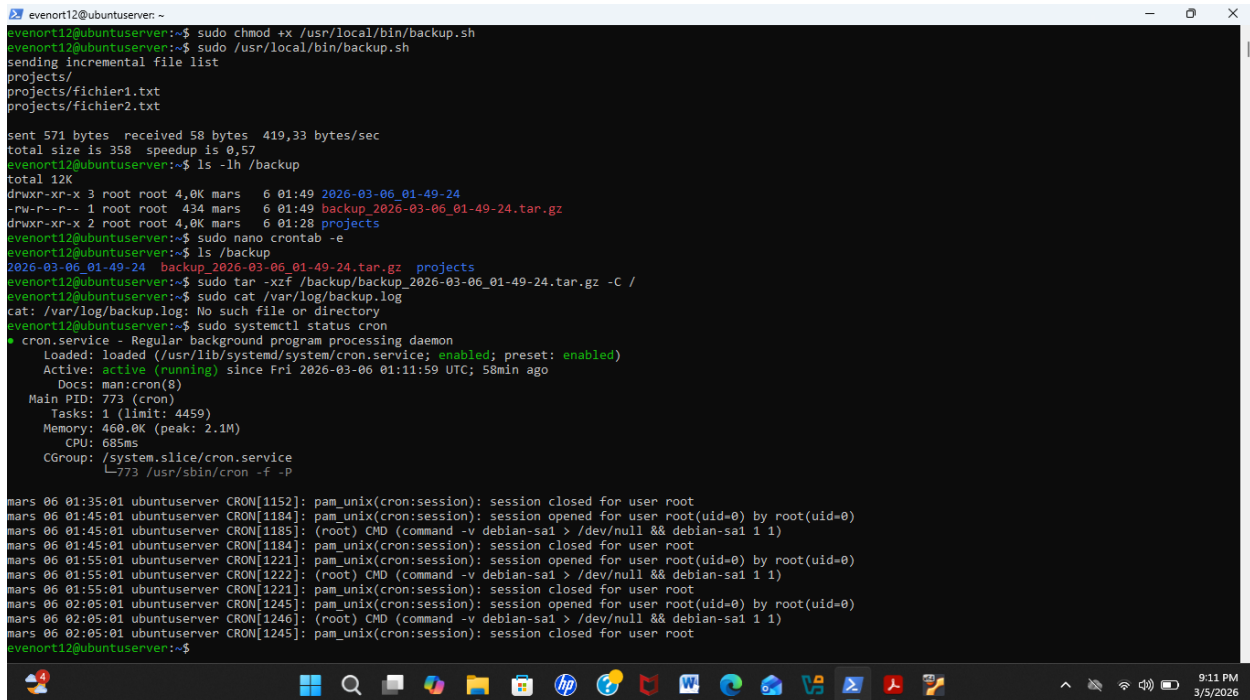


The screenshot shows a terminal window with the title 'evenort12@ubuntu: ~'. The user has executed several commands: `ls /home/devops/projects` (output: fichier2.txt), `sudo rsync -av /backup/projects/ /home/devops/projects` (output: sending incremental file list, ./, fichier1.txt), `ls /home/devops/projects` (output: fichier1.txt fichier2.txt), `sudo nano /usr/local/bin/backup.sh`, `sudo chmod +x /usr/local/bin/backup.sh`, `sudo /usr/local/bin/backup.sh` (output: sending incremental file list, projects/, projects/fichier1.txt, projects/fichier2.txt), `ls -lh /backup` (output: total 12K, drwxr-xr-x 3 root root 4,0K mars 6 01:49 2026-03-06_01-49-24, -rw-r--r-- 1 root root 434 mars 6 01:49 backup_2026-03-06_01-49-24.tar.gz, drwxr-xr-x 2 root root 4,0K mars 6 01:28 projects), `sudo nano crontab -e`, `ls /backup` (output: 2026-03-06_01-49-24 backup_2026-03-06_01-49-24.tar.gz projects), and `crontab -e`. The nano editor is open, editing the 'crontab' file. The system's taskbar is visible at the bottom of the window, showing various application icons and the system clock indicating 9:00 PM on 3/5/2026.

Figure 13: Sur cette image, l'automatisation de la sauvegarde avec cron a été réalisée correctement.

6. Test de restauration

```
sudo tar -xzf /backup/backup_2026-03-06_01-49-24.tar.gz -C /
sudo systemctl status cron
```



```
evenort12@ubuntu:~$ sudo chmod +x /usr/local/bin/backup.sh
evenort12@ubuntu:~$ sudo /usr/local/bin/backup.sh
sending incremental file list
projects/
projects/fichier1.txt
projects/fichier2.txt

sent 571 bytes  received 58 bytes  419,33 bytes/sec
total size is 358  speedup is 0,57
evenort12@ubuntu:~$ ls -lh /backup
total 12K
drwxr-xr-x 3 root root 4,0K mars 6 01:49 2026-03-06_01-49-24
-rw-r--r-- 1 root root 434 mars 6 01:49 backup_2026-03-06_01-49-24.tar.gz
drwxr-xr-x 2 root root 4,0K mars 6 01:28 projects
evenort12@ubuntu:~$ sudo nano crontab -e
evenort12@ubuntu:~$ ls /backup
2026-03-06_01-49-24  backup_2026-03-06_01-49-24.tar.gz  projects
evenort12@ubuntu:~$ sudo tar -xzf /backup/backup_2026-03-06_01-49-24.tar.gz -C /
evenort12@ubuntu:~$ sudo cat /var/log/backup.log
cat: /var/log/backup.log: No such file or directory
evenort12@ubuntu:~$ sudo systemctl status cron
● cron.service - Regular background program processing daemon
   Loaded: loaded (/usr/lib/systemd/system/cron.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-03-06 01:11:59 UTC; 50min ago
     Docs: man:cron(8)
    Main PID: 773 (cron)
      Tasks: 1 (limit: 4459)
     Memory: 460,0K (peak: 2.1M)
        CPU: 685ms
    CGroup: /system.slice/cron.service
            └─773 /usr/sbin/cron -f -P

mars 06 01:35:01 ubuntu:server CRON[1152]: pam_unix(cron:session): session closed for user root
mars 06 01:45:01 ubuntu:server CRON[1184]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
mars 06 01:45:01 ubuntu:server CRON[1185]: (root) CMD (command -v debian-sa1 > /dev/null && debian-sa1 1 1)
mars 06 01:45:01 ubuntu:server CRON[1184]: pam_unix(cron:session): session closed for user root
mars 06 01:55:01 ubuntu:server CRON[1221]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
mars 06 01:55:01 ubuntu:server CRON[1222]: (root) CMD (command -v debian-sa1 > /dev/null && debian-sa1 1 1)
mars 06 01:55:01 ubuntu:server CRON[1221]: pam_unix(cron:session): session closed for user root
mars 06 02:05:01 ubuntu:server CRON[1245]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
mars 06 02:05:01 ubuntu:server CRON[1246]: (root) CMD (command -v debian-sa1 > /dev/null && debian-sa1 1 1)
mars 06 02:05:01 ubuntu:server CRON[1245]: pam_unix(cron:session): session closed for user root
evenort12@ubuntu:~$
```

Figure 14: Sur cette image, le test de restauration et le service cron ont été réalisés avec succès.

Exercice 2 – Sauvegarde distante avec RSYNC + SSH

1. Architecture

Serveur principal (SRV-PROD)

Données critiques :

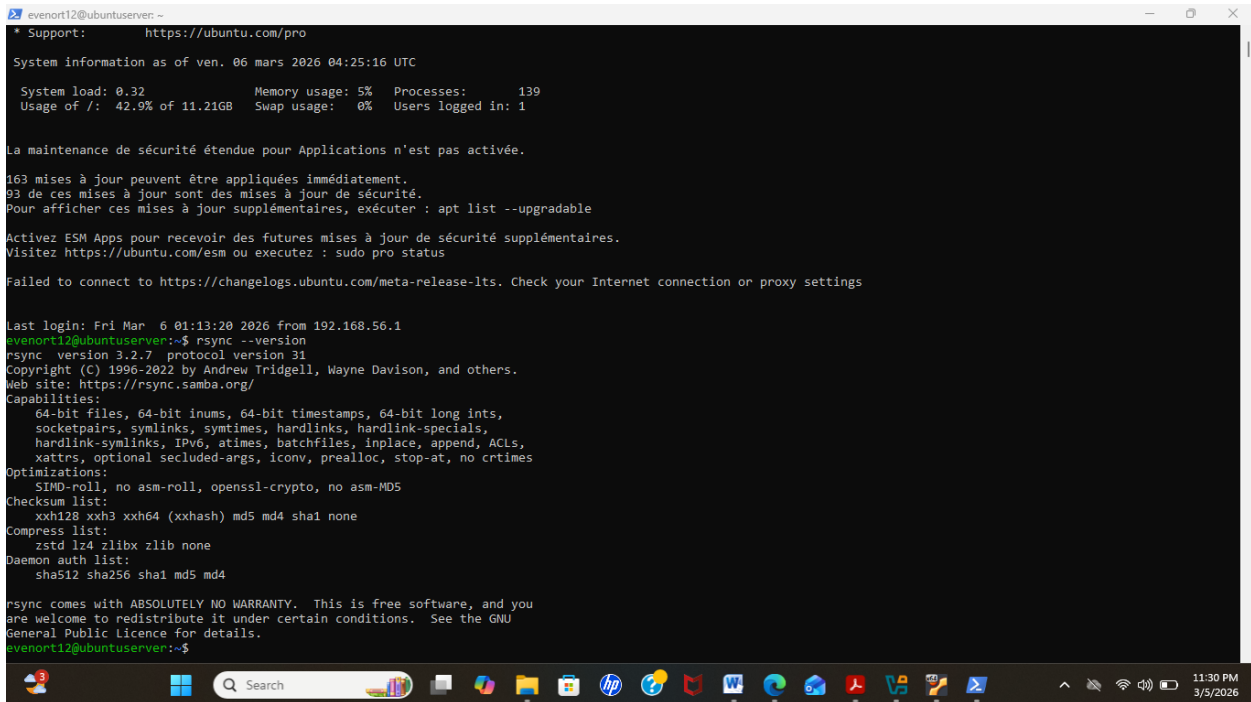
/var/www/

/etc/

/home/devops/projects

1.1 Partie 1 – Préparation du serveur de sauvegarde

a) Vérification de l'installation de rsync

A terminal window titled 'evenort12@ubuntu: ~' showing the output of the 'rsync --version' command. The output includes system information, security notices, and detailed rsync version and capabilities. The terminal window has a dark background and a standard Ubuntu desktop environment at the bottom with a taskbar and system tray.

```
evenort12@ubuntu: ~  
* Support: https://ubuntu.com/pro  
System information as of ven. 06 mars 2026 04:25:16 UTC  
System load: 0.32 Memory usage: 5% Processes: 139  
Usage of /: 42.9% of 11.21GB Swap usage: 0% Users logged in: 1  
  
La maintenance de sécurité étendue pour Applications n'est pas activée.  
163 mises à jour peuvent être appliquées immédiatement.  
93 de ces mises à jour sont des mises à jour de sécurité.  
Pour afficher ces mises à jour supplémentaires, exécuter : apt list --upgradable  
  
Activez ESM Apps pour recevoir des futures mises à jour de sécurité supplémentaires.  
Visitez https://ubuntu.com/esm ou exécutez : sudo pro status  
  
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings  
  
Last login: Fri Mar 6 01:13:20 2026 from 192.168.56.1  
evenort12@ubuntu: ~$ rsync --version  
rsync version 3.2.7 protocol version 31  
Copyright (C) 1996-2022 by Andrew Tridgell, Wayne Davison, and others.  
Web site: https://rsync.samba.org/  
Capabilities:  
64-bit files, 64-bit inums, 64-bit timestamps, 64-bit long ints,  
socketpairs, symlinks, symtimes, hardlinks, hardlink-specials,  
hardlink-symlinks, IPv6, atimes, batchfiles, inplace, append, ACLs,  
xattrs, optional secluded-args, iconv, prealloc, stop-at, no crtms  
Optimizations:  
SIMD-roll, no asm-roll, openssl-crypto, no asm-MD5  
Checksum list:  
xxh128 xxh3 xxh64 (xxhash) md5 md4 sha1 none  
Compress list:  
zstd lz4 zlib none  
Daemon auth list:  
sha512 sha256 sha1 md5 md4  
  
rsync comes with ABSOLUTELY NO WARRANTY. This is free software, and you  
are welcome to redistribute it under certain conditions. See the GNU  
General Public Licence for details.  
evenort12@ubuntu: ~$
```

Figure 15: Cette image montre que rsync est bel et bien installé.

3. Authentification SSH par clé

1.2 Créer une clé SSH : ssh-keygen

A) Envoi de la clé vers le serveur de sauvegarde

ssh-copy-id backup@192.168.56.123

```
evenort12@ubuntu:~$ ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/evenort12/.ssh/id_ed25519):
/home/evenort12/.ssh/id_ed25519 already exists.
Overwrite (y/n)? y
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/evenort12/.ssh/id_ed25519
Your public key has been saved in /home/evenort12/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:0iFyodgnumhlnk4iKloRiQ4tECOWdOyuzPFEm1GXB evenort12@ubuntu
The key's randomart image is:
+--[ED25519 256]--+
|
|o.
|++
|*+ . o o
|oo * .
|o + S .
|..+ o + @ + E
|o o o % o o
|..o + + .
|=.+. .o
|-----[SHA256]-----
evenort12@ubuntu:~$ cat .ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZ0I1NTU5AAAAIAAp12jUTPoildsYmIgoPAzFxVf4kCpM8Bj7wtuGv7 evenort12@ubuntu
evenort12@ubuntu:~$ ssh-copy-id backup@192.168.56.123
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/evenort12/.ssh/id_ed25519.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- If you are prompted now it is to install the new keys
backup@192.168.56.123's password:
mkdir: cannot create directory '.ssh': Permission denied
evenort12@ubuntu:~$ ssh backup@192.168.56.123
backup@192.168.56.123's password:
```

Figure 16 : Cette étape consiste à transférer la clé SSH vers le serveur de sauvegarde afin de permettre une connexion sécurisée sans mot de passe.

NB : J’ai exécuté cette opération manuellement via le serveur Ubuntu.

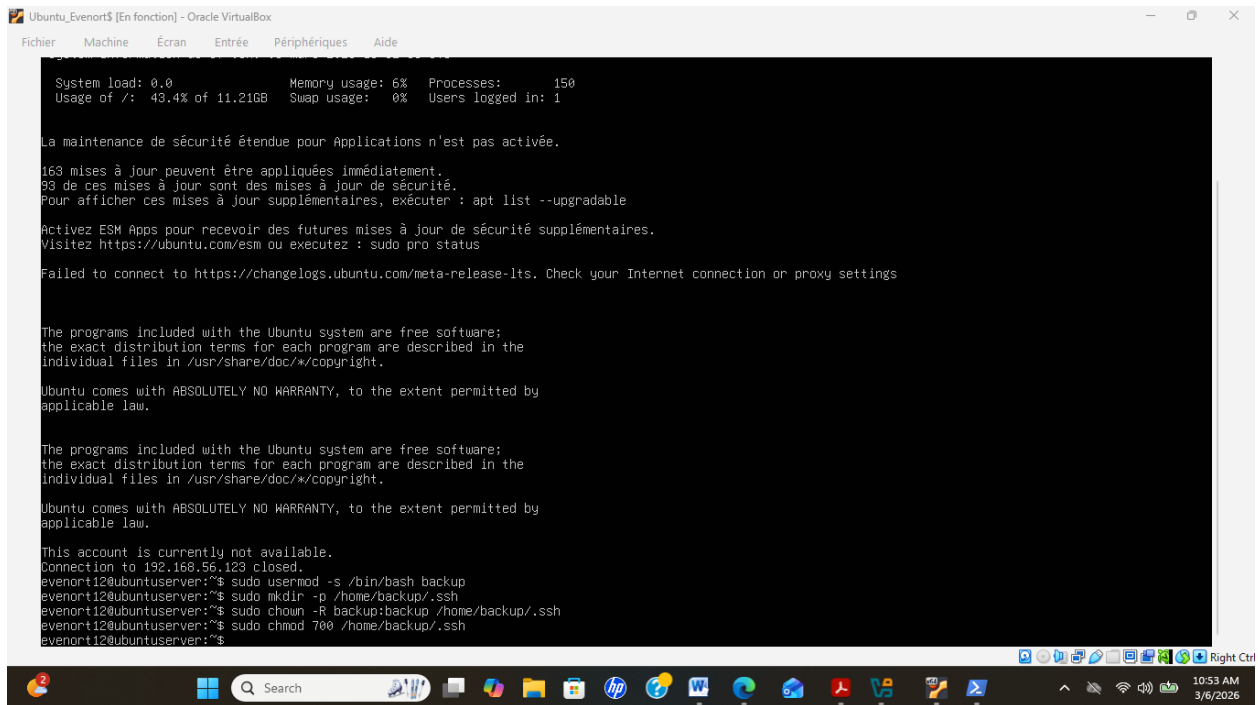
1.1 Création du dossier sur SRV-BACKUP

Sudo mkdir -p /home/backup/.ssh

1.2 Permissions

Sudo chown -R backup:backup /home/backup/.ssh

Sudo chmod 700 /home/backup/.ssh



```
System load: 0.0      Memory usage: 6%    Processes:   150
Usage of /: 43.4% of 11.21GB  Swap usage: 0%    Users logged in: 1

La maintenance de sécurité étendue pour Applications n'est pas activée.
163 mises à jour peuvent être appliquées immédiatement.
93 de ces mises à jour sont des mises à jour de sécurité.
Pour afficher ces mises à jour supplémentaires, exécuter : apt list --upgradable
Activez ESM Apps pour recevoir des futures mises à jour de sécurité supplémentaires.
Visitez https://ubuntu.com/esm ou exécutez : sudo pro status
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

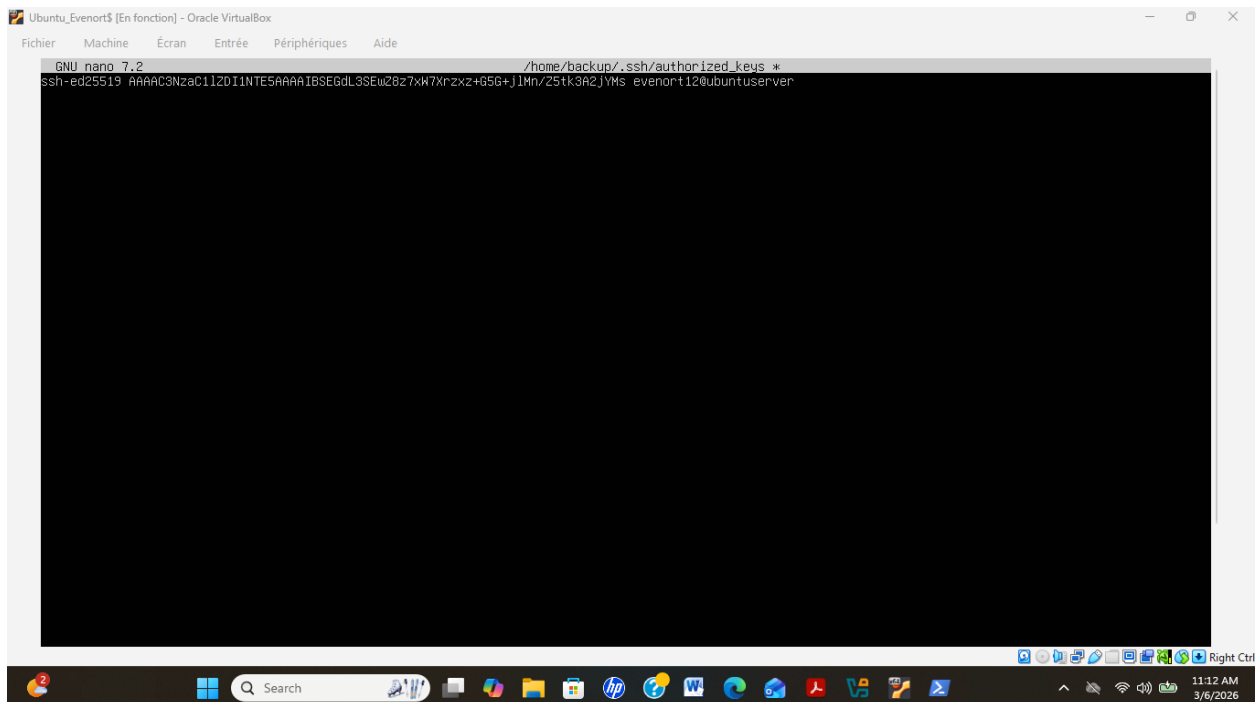
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

This account is currently not available.
Connection to 192.168.56.123 closed.
evenort12@ubuntu:~$ sudo usermod -s /bin/bash backup
evenort12@ubuntu:~$ sudo mkdir -p /home/backup/.ssh
evenort12@ubuntu:~$ sudo chown -R backup:backup /home/backup/.ssh
evenort12@ubuntu:~$ sudo chmod 700 /home/backup/.ssh
evenort12@ubuntu:~$
```

Figure 17: Cette étape consiste à créer un dossier dédié sur le serveur de sauvegarde (SRV-BACKUP) pour y stocker les fichiers sauvegardés.

1.3 Ajouter : `sudo nano /home/backup/.ssh/authorized_keys`



```
GNU nano 7.2 /home/backup/.ssh/authorized_keys *
ssh-ed25519 AAAAC3NzaC1l2D1lINTE5AAAA1BSEGdL3SEw28z7xM7Xr2xz+65G+J1Mn/251K3A2JYMs evenort12@ubuntu:~$
```

Figure 18 : Sur cette image, le contenu (clé SSH) du fichier a été ajouté afin de permettre une connexion sécurisée au serveur de sauvegarde.

2. Connexion a un server distant via SSH

3. Tester la connexion sans mot de passe

```
evenort12@ubuntu:~$ ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/evenort12/.ssh/id_ed25519):
/home/evenort12/.ssh/id_ed25519 already exists.
Overwrite (y/n)? y
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/evenort12/.ssh/id_ed25519
Your public key has been saved in /home/evenort12/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:oiFyx68gnu8hlnk4iKloRiQ4tECOMdLoyuzPFEm1GX8 evenort12@ubuntu:~
The key's randomart image is:
++[ED25519 256]--+
|oo.
|+=.
|*+. .o o
|+oo * . .
|+o + $
|+ o + @ + E
|o o % o o
|+o + =
|+ .o
+----[SHA256]-----+
evenort12@ubuntu:~$ cat .ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIARpiZjUTPqIidsYw8g0PAzFxVf4kCpWRH0JJ7mtwGv7 evenort12@ubuntu:~
evenort12@ubuntu:~$ ssh-copy-id backup@192.168.56.123
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/evenort12/.ssh/id_ed25519.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
backup@192.168.56.123's password:
mkdir: cannot create directory '/ssh': Permission denied
evenort12@ubuntu:~$ ssh backup@192.168.56.123
backup@192.168.56.123's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-71-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of ven. 06 mars 2026 16:22:01 UTC

System load: 0.13          Memory usage: 7%    Processes:    152
Usage of /:  43.4% of 11.21GB Swap usage: 0%    Users logged in: 2
```

```
evenort12@ubuntu:~$ ssh backup@192.168.56.123
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-71-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of ven. 06 mars 2026 16:22:01 UTC

System load: 0.13          Memory usage: 7%    Processes:    152
Usage of /:  43.4% of 11.21GB Swap usage: 0%    Users logged in: 2

La maintenance de sécurité étendue pour Applications n'est pas activée.

163 mises à jour peuvent être appliquées immédiatement.
93 de ces mises à jour sont des mises à jour de sécurité.
Pour afficher ces mises à jour supplémentaires, exécuter : apt list --upgradable

Activez ESM Apps pour recevoir des futures mises à jour de sécurité supplémentaires.
Visitez https://ubuntu.com/esm ou exécutez : sudo pro status

Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

The programs included with the Ubuntu system are free software;
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individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

Last login: Fri Mar  6 16:18:07 2026 from 192.168.56.123
backup@ubuntu:~$
```



```
evenort12@ubuntu:~$ ssh backup@192.168.56.123
backup@ubuntu:~$ ssh backup@192.168.56.123
The authenticity of host '192.168.56.123 (192.168.56.123)' can't be established.
ED25519 key fingerprint is SHA256:hZsPxbTDIaGy4z751X+EY151wC3UYNHkng2/RPnrDz0.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Could not create directory '/var/backups/.ssh' (Permission denied).
Failed to add the host to the list of known hosts (/var/backups/.ssh/known_hosts).
backup@192.168.56.123's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-71-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of ven. 06 mars 2026 16:38:09 UTC

System load: 0.06      Memory usage: 7%      Processes:   153
Usage of /: 43.4% of 11.21GB  Swap usage: 0%      Users logged in: 2

La maintenance de sécurité étendue pour Applications n'est pas activée.

163 mises à jour peuvent être appliquées immédiatement.
93 de ces mises à jour sont des mises à jour de sécurité.
Pour afficher ces mises à jour supplémentaires, exécuter : apt list --upgradable

Activez ESM Apps pour recevoir des futures mises à jour de sécurité supplémentaires.
Visitez https://ubuntu.com/esm ou exécutez : sudo pro status

Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
```

Figure 19, 20, 21: Ces images illustrent que la connexion à un serveur distant via SSH a été établie avec succès (voir pages 16 à 17).

4. Sauvegarde distante avec rsync

rsync -avz /home/devops/projects [backup@192.168.56.123:/backup/](ssh://backup@192.168.56.123:/backup/)

5. Automatisation avec Cron

crontab -e

Ajouter :

**0 23 * * * rsync -avz /home/devops/projects
[backup@192.168.56.123:/backup/](ssh://backup@192.168.56.123:/backup/)**

```
evenort12@ubuntu: ~  
[1/2]  
0 23 * * * /usr/local/bin/backup.sh >> /var/log/backup.log 2>&1  
0 23 * * * rsync -avz /home/devops/projects backup@192.168.56.123:/backup/
```

Figure 22: Cette image illustre que le contenu du fichier cron a été ajouté afin d’automatiser les sauvegardes avec rsync.

6. Vérification: sudo systemctl status cron

```
evenort12@ubuntu: ~  
Visitez https://ubuntu.com/esm ou exécutez : sudo pro status  
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings  
  
Last login: Fri Mar 6 14:48:01 2026 from 192.168.56.1  
evenort12@ubuntu:~$ rsync -avz /home/devops/projects backup@192.168.56.123:/backup/  
backup@192.168.56.123's password:  
sending incremental file list  
  
sent 120 bytes received 17 bytes 21.08 bytes/sec  
total size is 358 speedup is 2.61  
evenort12@ubuntu:~$ crontab -e  
-e: No such file or directory  
evenort12@ubuntu:~$ nanocrontab -e  
nanocrontab: command not found  
evenort12@ubuntu:~$ nano crontab -e  
evenort12@ubuntu:~$  
evenort12@ubuntu:~$ sudo nano crontab -e  
[sudo] password for evenort12:  
evenort12@ubuntu:~$ sudo systemctl status cron  
● cron.service - Regular background program processing daemon  
   Loaded: loaded (/usr/lib/systemd/system/cron.service; enabled; preset: enabled)  
   Active: active (running) since Fri 2026-03-06 14:11:44 UTC; 3h 18min ago  
     Docs: man:cron(8)  
    Main PID: 844 (cron)  
      Tasks: 1 (limit: 4450)  
    Memory: 456.0K (peak: 2.3M)  
       CPU: 1.211s  
    CGroup: /system.slice/cron.service  
            └─844 /usr/sbin/cron -f -P  
  
mars 06 17:05:01 ubuntu:server CRON[3084]: pam_unix(cron:session): session closed for user root  
mars 06 17:09:01 ubuntu:server cron[844]: (backup) RELOAD (crontabs/backup)  
mars 06 17:15:01 ubuntu:server CRON[3045]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)  
mars 06 17:15:01 ubuntu:server CRON[3046]: (root) CMD (command -v debian-sa1 > /dev/null && debian-sa1 1 1)  
mars 06 17:15:01 ubuntu:server CRON[3045]: pam_unix(cron:session): session closed for user root  
mars 06 17:17:01 ubuntu:server CRON[3048]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)  
mars 06 17:17:01 ubuntu:server CRON[3048]: pam_unix(cron:session): session closed for user root  
mars 06 17:20:01 ubuntu:server cron[844]: (backup) RELOAD (crontabs/backup)  
mars 06 17:25:01 ubuntu:server CRON[3135]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)  
mars 06 17:25:01 ubuntu:server CRON[3135]: pam_unix(cron:session): session closed for user root  
evenort12@ubuntu:~$
```

Figure 23: Cette image montre les lignes de commande utilisées et confirme que le service cron fonctionne correctement et de

```
evenort12@ubuntu:~$ nano crontab -e
-e: No such file or directory
evenort12@ubuntu:~$ nano crontab -e
nanocrontab: command not found
evenort12@ubuntu:~$ nano crontab -e
evenort12@ubuntu:~$ sudo nano crontab -e
[sudo] password for evenort12:
evenort12@ubuntu:~$ sudo systemctl status cron
• cron.service - Regular background program processing daemon
   Loaded: loaded (/usr/lib/systemd/system/cron.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-03-06 14:11:44 UTC; 3h 18min ago
     Docs: man:cron(8)
    Main PID: 844 (cron)
      Tasks: 1 (limit: 4450)
    Memory: 456.0K (peak: 2.3M)
       CPU: 1.211s
    CGroup: /system.slice/cron.service
            └─844 /usr/sbin/cron -f -P

mars 06 17:05:01 ubuntu: CRON[3004]: pam_unix(cron:session): session closed for user root
mars 06 17:09:01 ubuntu: cron[844]: (backup) RELOAD (crontabs/backup)
mars 06 17:15:01 ubuntu: CRON[3045]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
mars 06 17:15:01 ubuntu: CRON[3046]: (root) CMD (command -v debian-sa1 > /dev/null && debian-sa1 1 1)
mars 06 17:15:01 ubuntu: CRON[3045]: pam_unix(cron:session): session closed for user root
mars 06 17:17:01 ubuntu: CRON[3048]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
mars 06 17:17:01 ubuntu: CRON[3048]: pam_unix(cron:session): session closed for user root
mars 06 17:20:01 ubuntu: cron[844]: (backup) RELOAD (crontabs/backup)
mars 06 17:25:01 ubuntu: CRON[3135]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
mars 06 17:25:01 ubuntu: CRON[3135]: pam_unix(cron:session): session closed for user root
evenort12@ubuntu:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:64:2c:05 brd ff:ff:ff:ff:ff:ff
   inet 192.168.56.123/24 metric 100 brd 192.168.56.255 scope global dynamic enp0s3
       valid_lft 475sec preferred_lft 475sec
   inet6 fe80::a00:27ff:fe64:2c05/64 scope link
       valid_lft forever preferred_lft forever
evenort12@ubuntu:~$
```

Figure 24: Cette image montre les lignes de commande utilisées et confirme que le service cron fonctionne correctement, permettant de s'assurer que les tâches planifiées s'exécutent comme prévu.

Questions

1. Différence entre tar et rsync

- **tar** : crée une archive compressée d'un dossier ou fichier.
- **rsync** : synchronise les fichiers et copie seulement les modifications (incrémental).

2. Différence entre scp et rsync

- **scp** : copie les fichiers via SSH.
- **rsync** : copie les fichiers mais optimise le transfert en envoyant seulement les changements.

3. Pourquoi automatiser les sauvegardes ?

R) Pour éviter la perte de données et garantir que les sauvegardes sont réalisées régulièrement sans intervention manuelle.

4. Où sont stockées les tâches cron de l'utilisateur ?

Dans le fichier :

`/var/spool/cron/crontabs/`

5. Comment planifier une sauvegarde tous les jours à 23h ?

Avec la ligne cron suivante : `0 23 * * * commande`

Conclusion

La mise en place d'une stratégie de sauvegarde est essentielle pour protéger les données d'une entreprise. Grâce à **rsync**, **SSH** et **cron**, il est possible de créer une solution de sauvegarde automatique, sécurisée et efficace.